

Key words And Terms

natural light

自然光

polarization

偏振

plane-polarized light

线偏振光

partially polarized light

部分偏振光

polaroid

偏振片

polarizer

偏振器

Malus law

马吕斯定律

Brewster angle

布儒斯特角



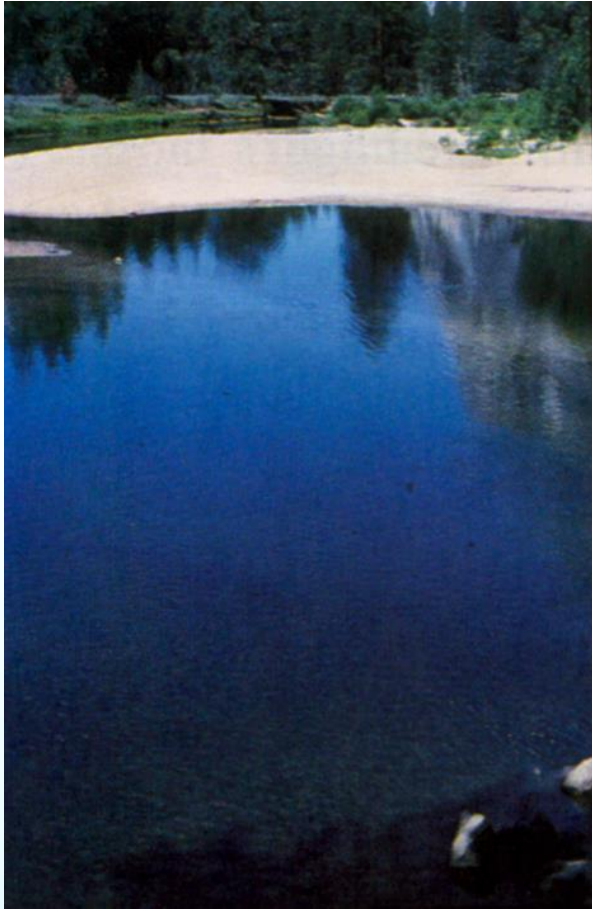
Chapter 20 Polarization

20.1 Polarization of Light

20.2 Malus law

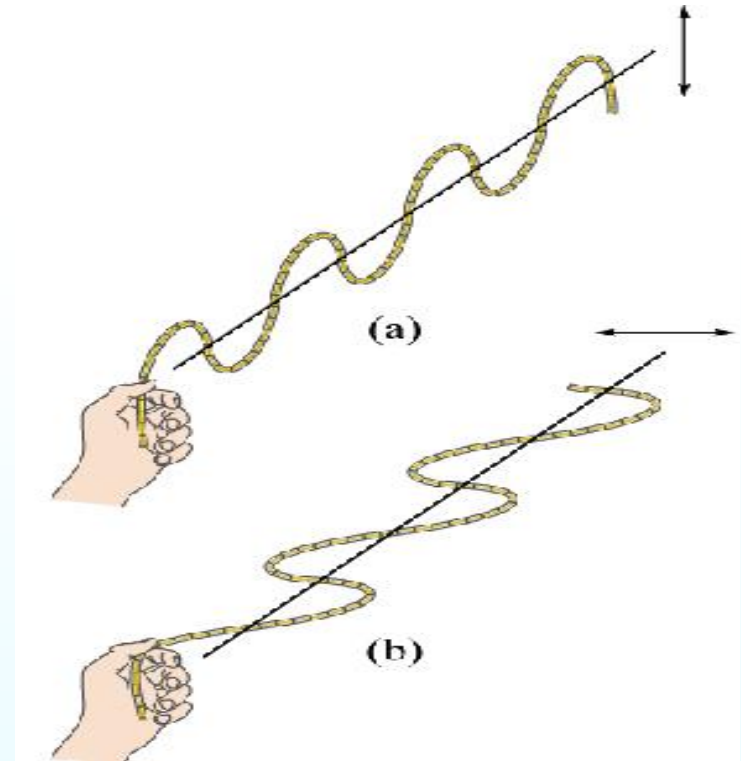
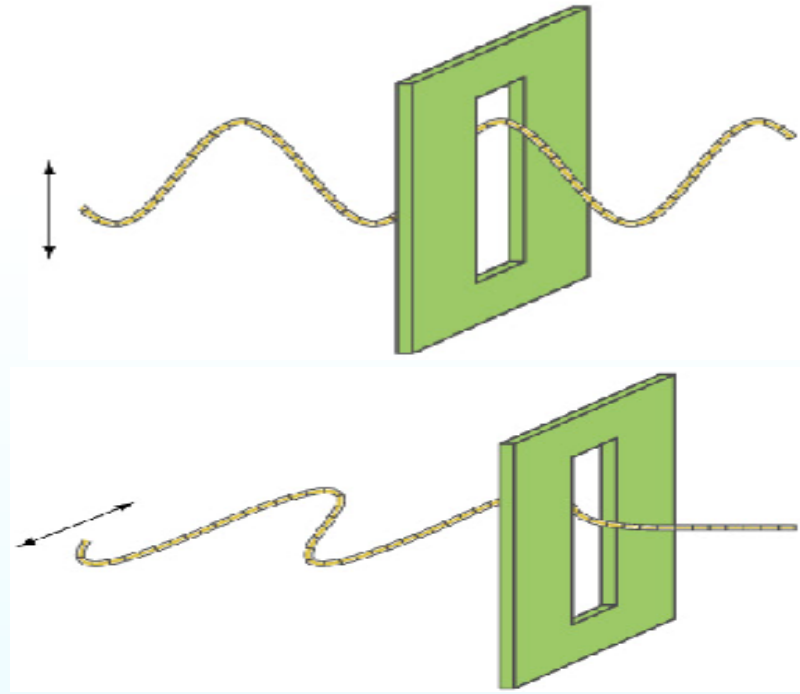
***20.3 Polarization by Reflection**

Chapter 20 Polarization



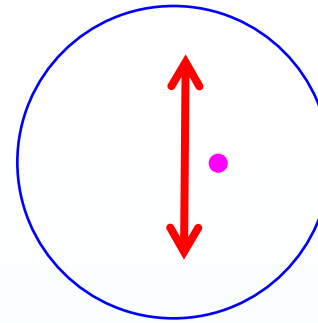
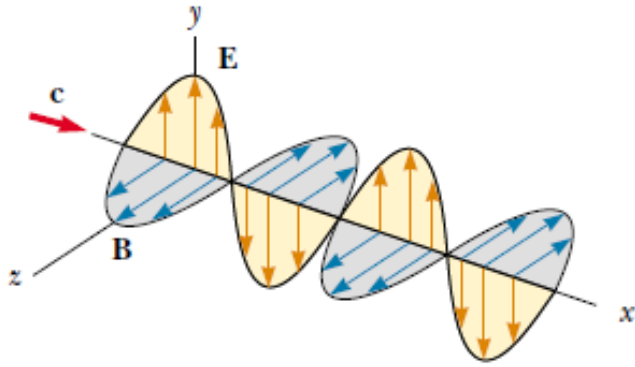
(a) Without polarizer; (b) Using a polarizer which is adjusted to absorb most of the polarized light reflected from the water's surface, and any fish to be seen more readily.

20.1 Polarization of Light



Polarization can exist *only* for *transverse waves* , and not for longitudinal waves.

偏振：振动方向对于波的传播方向不具有柱对称性叫偏振。



**Head-on view
of the plane
of oscillation**

Polarization of light proves that **light is transverse wave** (光的偏振证明了光的横波性).

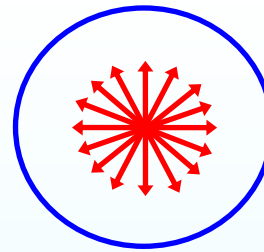
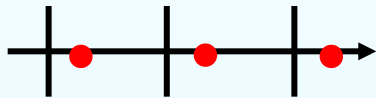
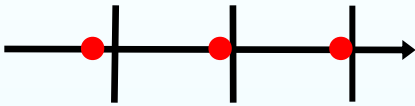
The direction of polarization in a EM wave is taken as the direction of the **electric field vector.**

1. Polarization of Light and representation

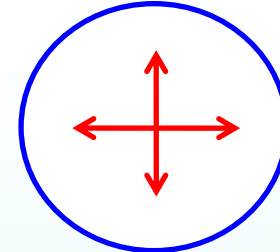
(1) Unpolarized light

The source of light has oscillations in many planes.

Denoting Method:



没有优势方向



自然光的分解

Oscillation of the electric field vectors

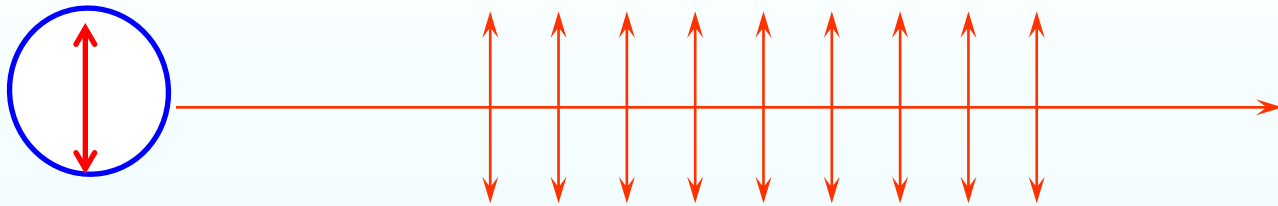
Unpolarized light consists of light with *random* directions of polarization and resolve the directions into **two** mutually perpendicular directions.

An ordinary incandescent lightbulb emits unpolarized light, as does the Sun.

(2) Plane-polarized Light (linearly polarized light)

Its field oscillations are in a fixed plane (direction) 光矢量
只在一个固定平面内沿一个固定方向振动的光叫**线偏振光**.

Denoting Method:



Component parallel to page

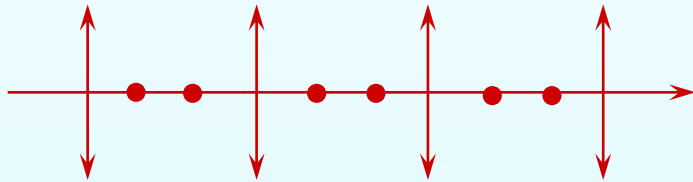


Component perpendicular to page

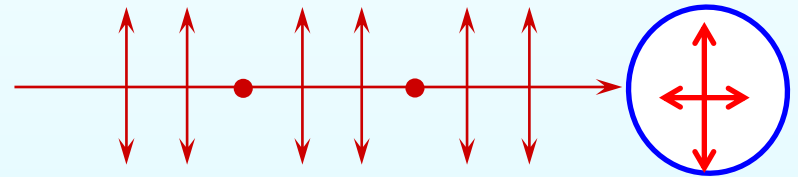
(3) Partially Polarized Light (部分偏振光)

Its field oscillations are neither completely random nor parallel to a single axis (the electric fields oscillating along one direction have **greater possibility than those oscillating along other ones**).

部分偏振光可分解为两束振动方向相互垂直的、不等幅的、不相干的线偏振光。



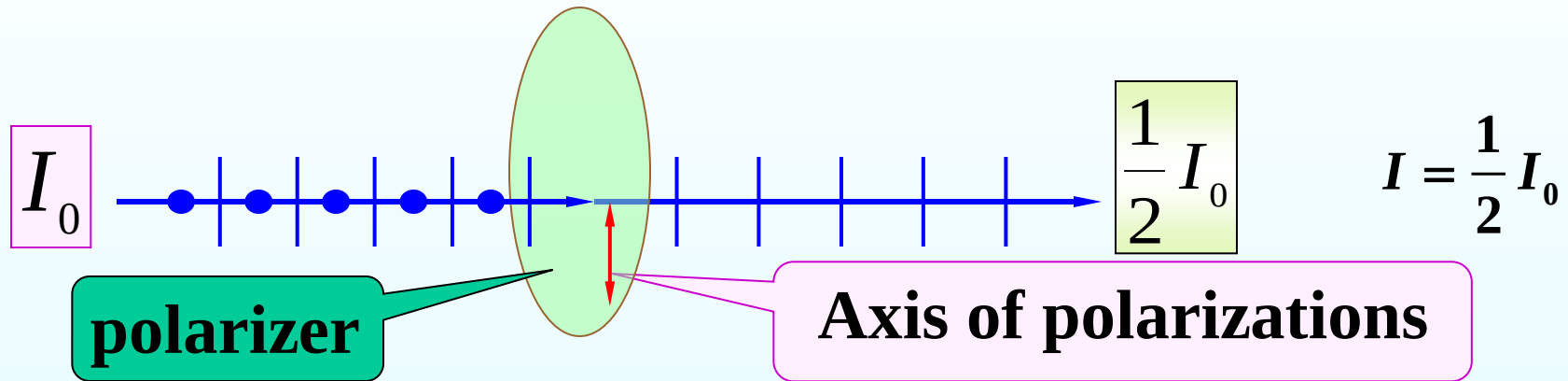
平行板面的光振动较强



垂直板面的光振动较强

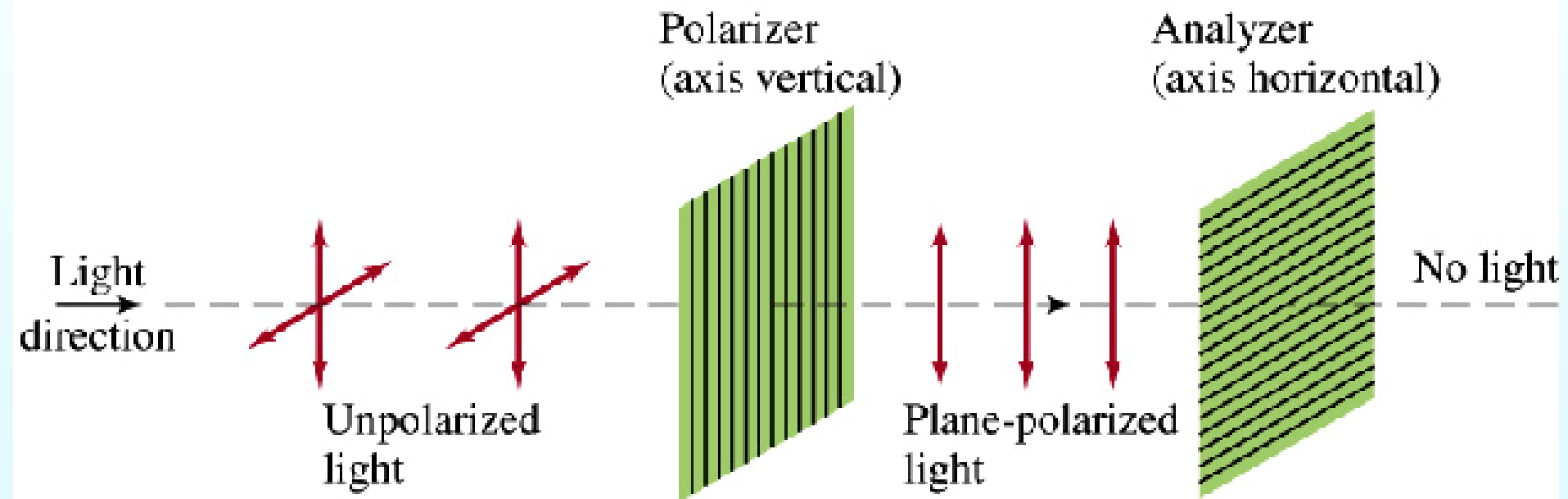
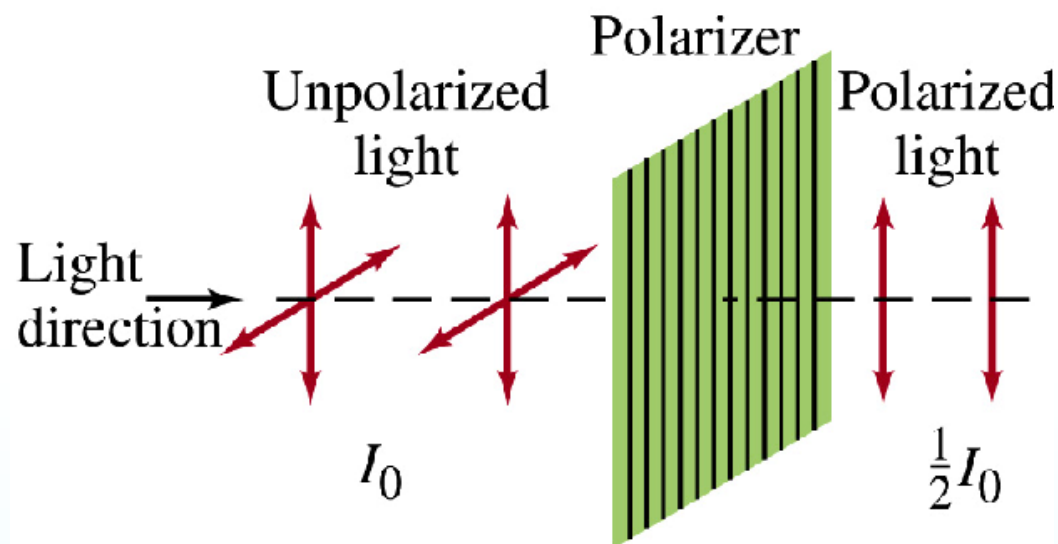
2. Polarizer and analyzer (起偏, 检偏)

Polaroid sheet(偏振片): A polaroid sheet allows the orientation of polarization to pass through nearly undiminished and the direction is called the **axis of the polaroid** (偏振化方向), whereas a perpendicular polarization is absorbed almost completely.



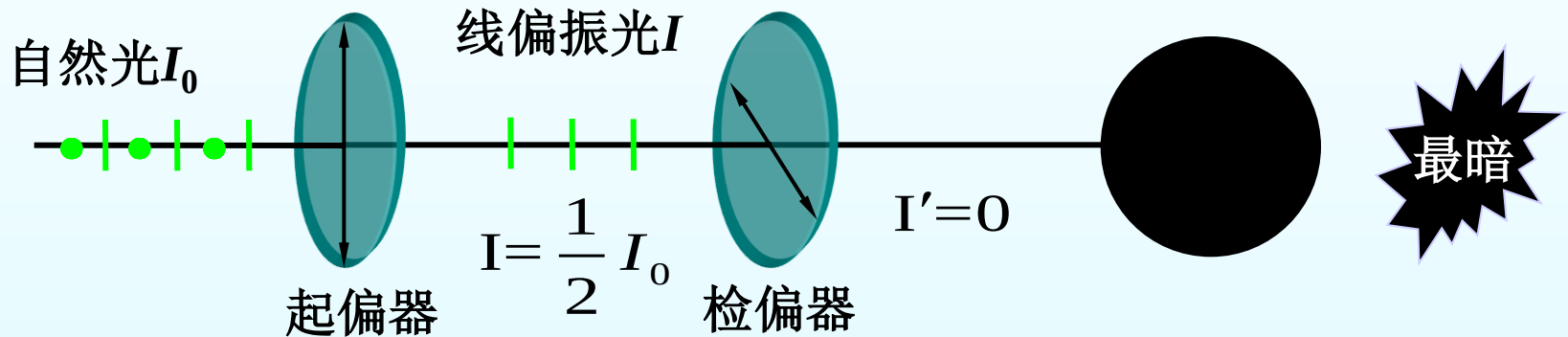
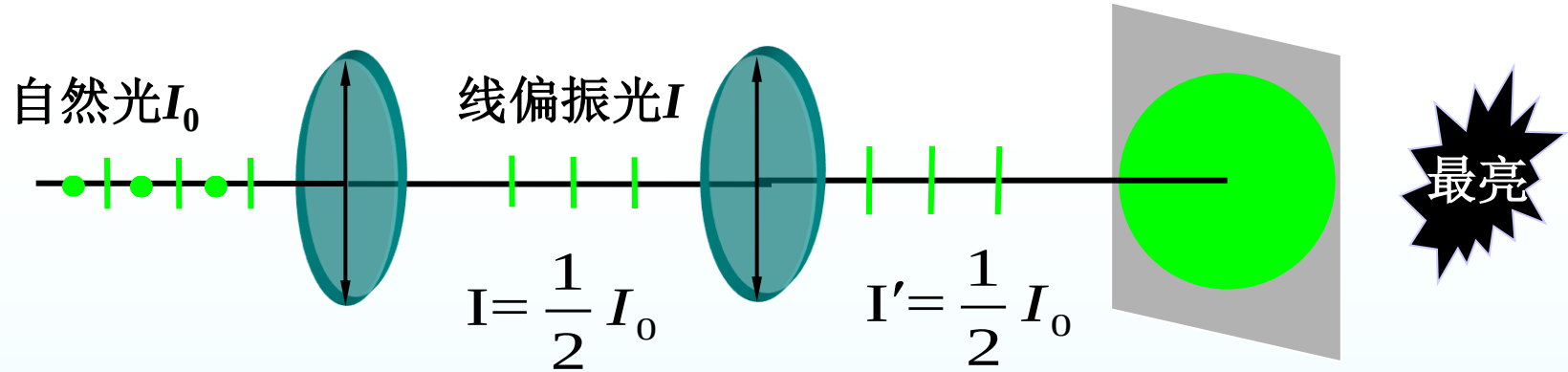
Polarizer (起偏): An unpolarized beam can be thought of as two perpendicular plane-polarized beams of equal magnitude. When unpolarized light passes through a polarizer, one of the components is eliminated.

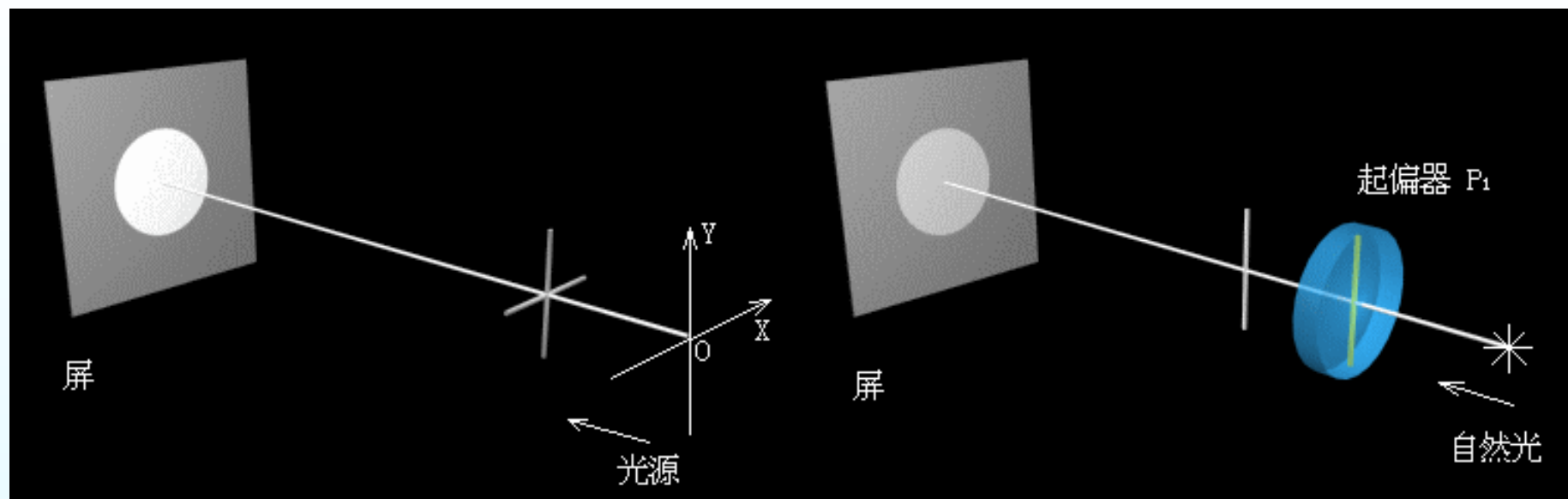
从自然光获得偏振光



Analyzer (检偏)

Plane-polarized Light





Analyzer

(1) Unpolarized light $I = \frac{1}{2} I_0$

The intensity of light transmitted does **not change**.

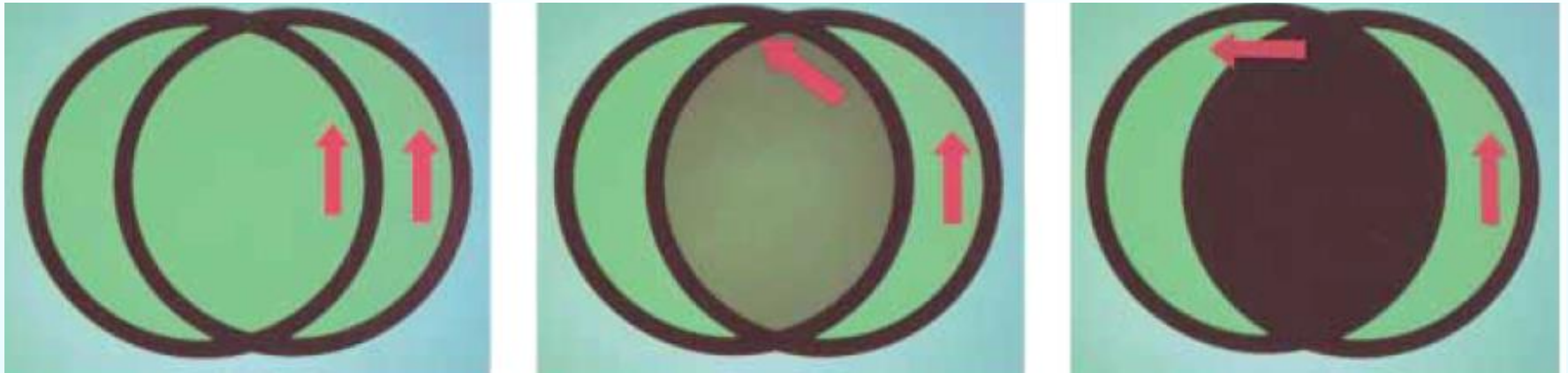
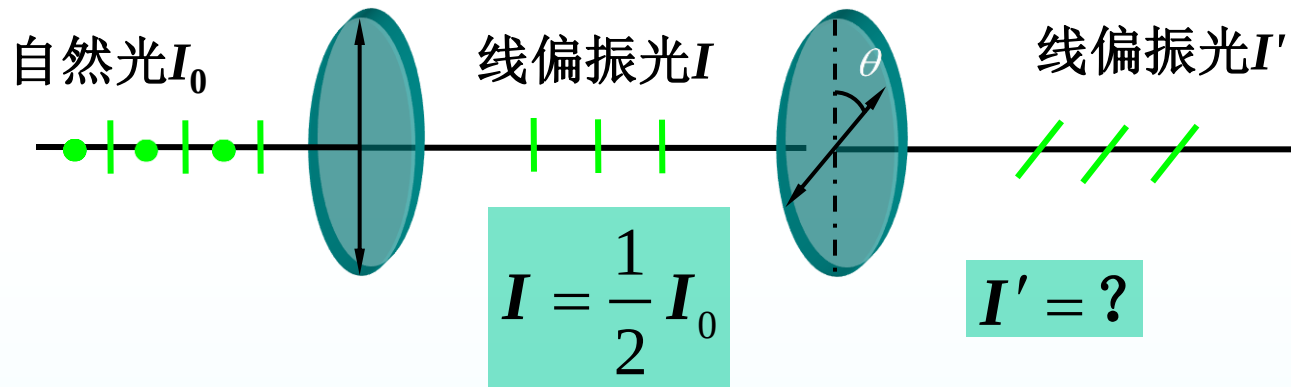
(2) Plane-polarized Light (linearly polarized light)

The intensity of light transmitted is **variable**, and the minimum is **zero**.

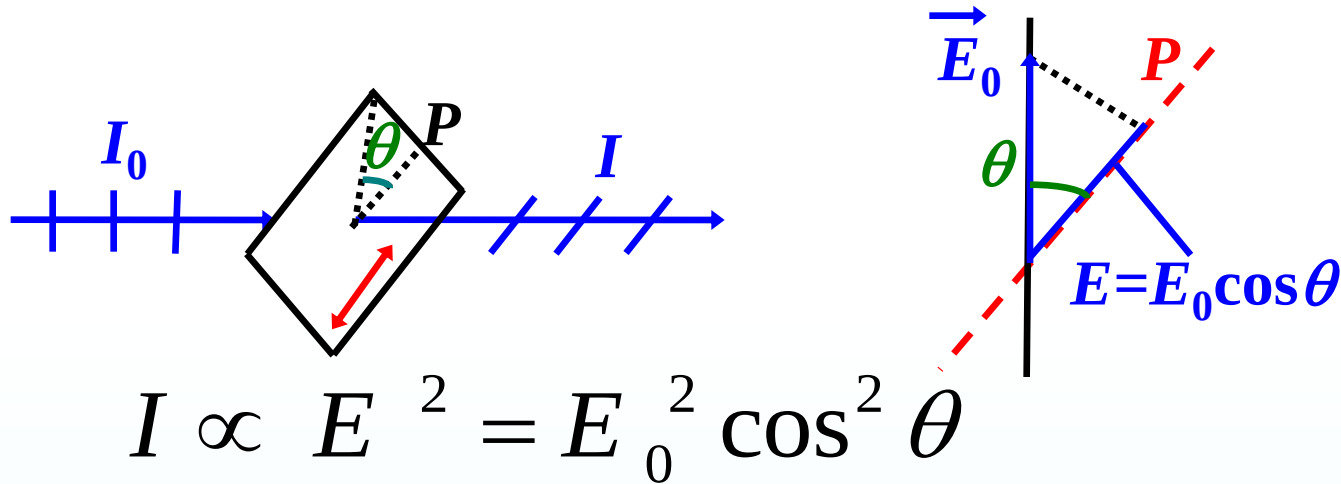
(3) Partially Polarized Light (部分偏振光)

The intensity of light transmitted is **variable**, and the minimum can **not be zero**.

20.2 Malus law (马吕斯定律)(1808):



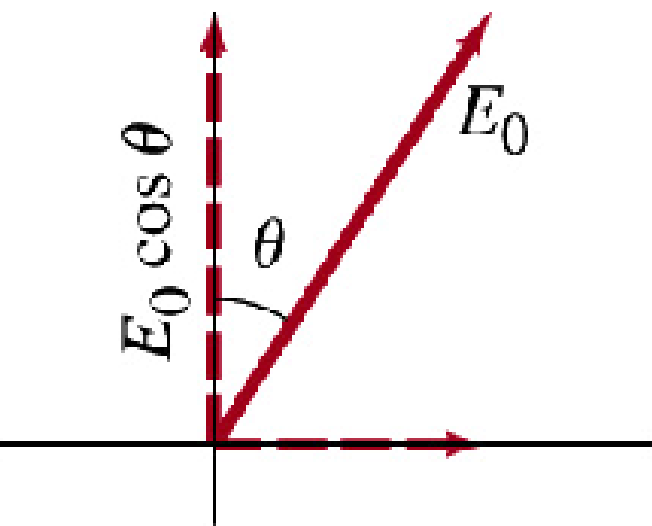
The intensity of light transmitted two polarizers depends on the *relative* orientation of their transmission axes.



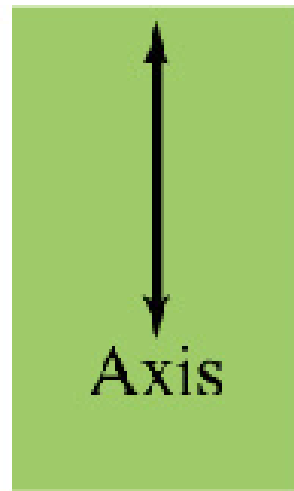
$$I = I_0 \cos^2 \theta \quad \text{—— Malus law (1809)}$$

Let us call this the cosine-squared rule, we can use it *only* when the light reaching a polarizing sheet is already *polarized*.

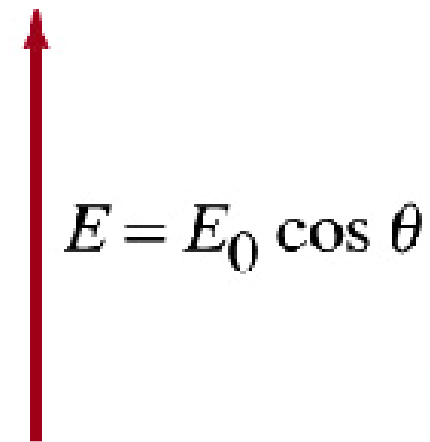
$$\theta = 0, \quad I = I_{\max} = I_0; \quad \theta = \frac{\pi}{2}, \quad I = 0 \quad \text{extinction}$$



Incident beam polarized
at angle θ to the vertical
and has amplitude E_0 .



Vertical
Polaroid



Transmitted wave

 **Note:**

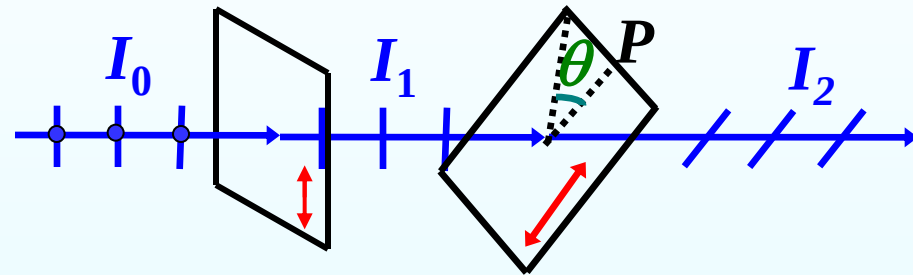
各偏振片只允许和自己的偏振
化方向相同的偏振光透过。

Example Unpolarized light passes through two polaroids; the axis of one is vertical and that of the other is at 60° to the vertical. What is the orientation and **intensity** of the **transmitted light**?

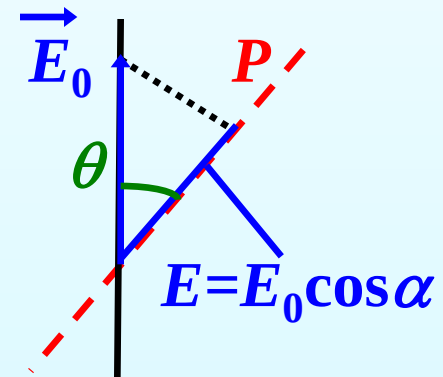
Solution: The 1st polaroid eliminates half the light, so

$$I = \frac{1}{2} I_0 \quad \text{One-half rule}$$

$$I_2 = I_1 (\cos^2 60^\circ) = \frac{1}{4} I_1 = \frac{1}{8} I_0$$



The light has an intensity one-eighth that of the original and is plane-polarized at a 60° angle to the vertical.



Example A system of three polarizing sheets in the path of initially unpolarized light as shown in the Fig.. What fraction of the initial intensity I_0 of the light emerges from the three-sheet system, and in which direction is that emerging light polarized?

Solution:

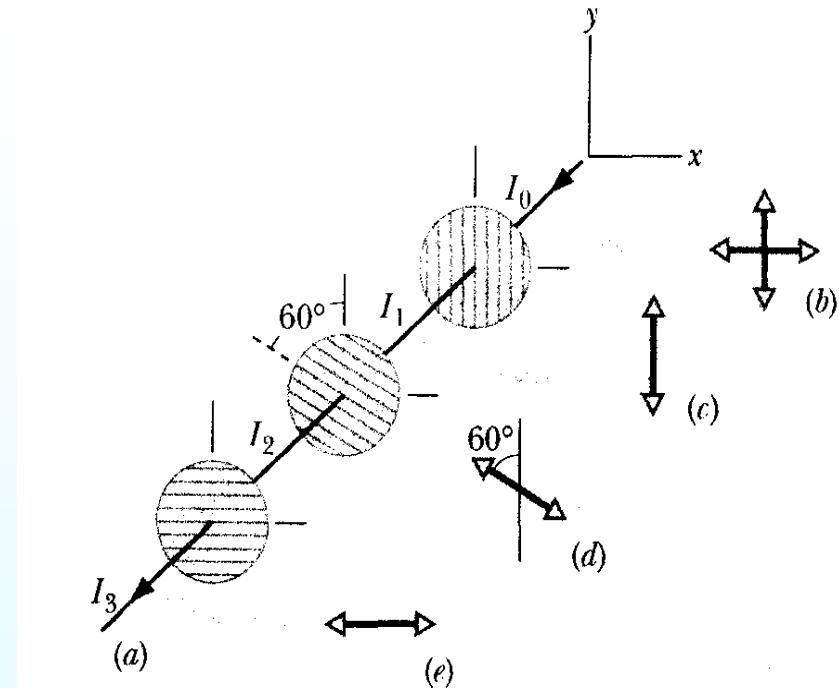
$$I_1 = \frac{1}{2} I_0 \quad I_2 = I_1 \cos^2 60^\circ$$

$$I_3 = I_2 \cos^2 30^\circ$$

$$I_3 = I_2 \cos^2 30^\circ = \left(\frac{I_0}{2}\right) \cos^2 60^\circ \cos^2 30^\circ = 0.094 I_0$$

$$\frac{I_3}{I_0} = 0.094$$

Parallel to the x axis



Example

At **what angle** should the axes of two polaroids be placed so as to reduce the intensity of the incident unpolarized light to $1/4$?

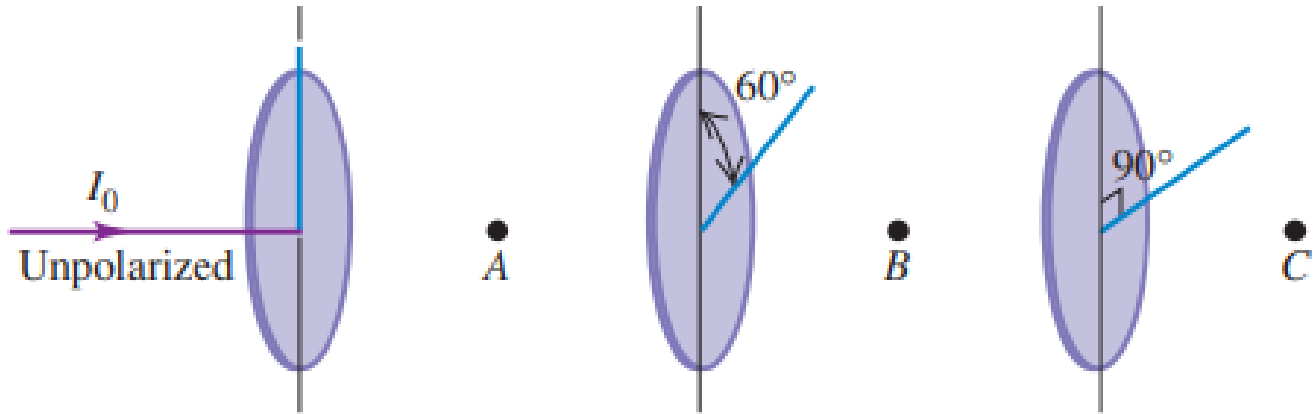
Solution:

$$I = I_0 \times \frac{1}{2} \times \cos^2 \theta = \frac{1}{4} I_0$$

$$\theta = \arccos \frac{\sqrt{2}}{2}$$

$$\theta = \frac{\pi}{4}$$

A beam of unpolarized light of intensity I_0 passes through a series of ideal polarizing filters with their polarizing directions turned to various angles as shown in Fig.. What is the light intensity (in terms of) at point C ?



- A. $I_0/32$
- B. $3I_0/32$
- C. $I_0/16$
- D. $3I_0/16$
- E. $I_0/8$

Example **Four** polarizers are placed in succession with their axes vertical, at 30° to the vertical, at 60° to the vertical, and at 90° to the vertical. (a) Calculate what fraction of the incident unpolarized light is transmitted by the four polarizers. (b) Can the transmitted light be **decreased** by removing one of the polarizers? If so, which one? (c) Can the transmitted light intensity be extinguished by removing polarizers? If so, Which one(s)?

Solution:

$$(a) \quad I = \frac{1}{2} I_0 \cos^2 30^\circ \cos^2 30^\circ \cos^2 30^\circ = \frac{27}{128} I_0 (0.21 I_0)$$

(b) Remove the second or the third one.

(c) Remove the second and third ones.

一束光是自然光和线偏振光的混合，当它通过一偏振片时，发现透射光的强度取决于偏振片的取向，其强度变化5倍，求入射光中自然光与线偏振光的光强比值为



(A) $\frac{1}{2}$

(B) $\frac{1}{5}$

(C) $\frac{1}{3}$

(D) $\frac{2}{3}$

一束光是自然光和线偏振光的混合，当它通过一偏振片时，发现透射光的强度取决于偏振片的取向，其强度变化5倍，求入射光中自然光与线偏振光的光强比值为



解： 设入射光强度： I_1 ； 自然光强度： I_2 ；

透射光最大强度 $I_{\max} = \frac{1}{2} I_1 + I_2$

透射光最小强度 $I_{\min} = \frac{1}{2} I_1$

$$I_{\max} = 5I_{\min} \quad \frac{\frac{1}{2} I_1 + I_2}{\frac{1}{2} I_1} = 5 \quad \rightarrow I_1 = \frac{1}{2} I_2$$

Describe how to rotate the plane of polarization of a plane-polarized beam of light by 90° and produce only a **10 percent loss on intensity, using polarizers.**



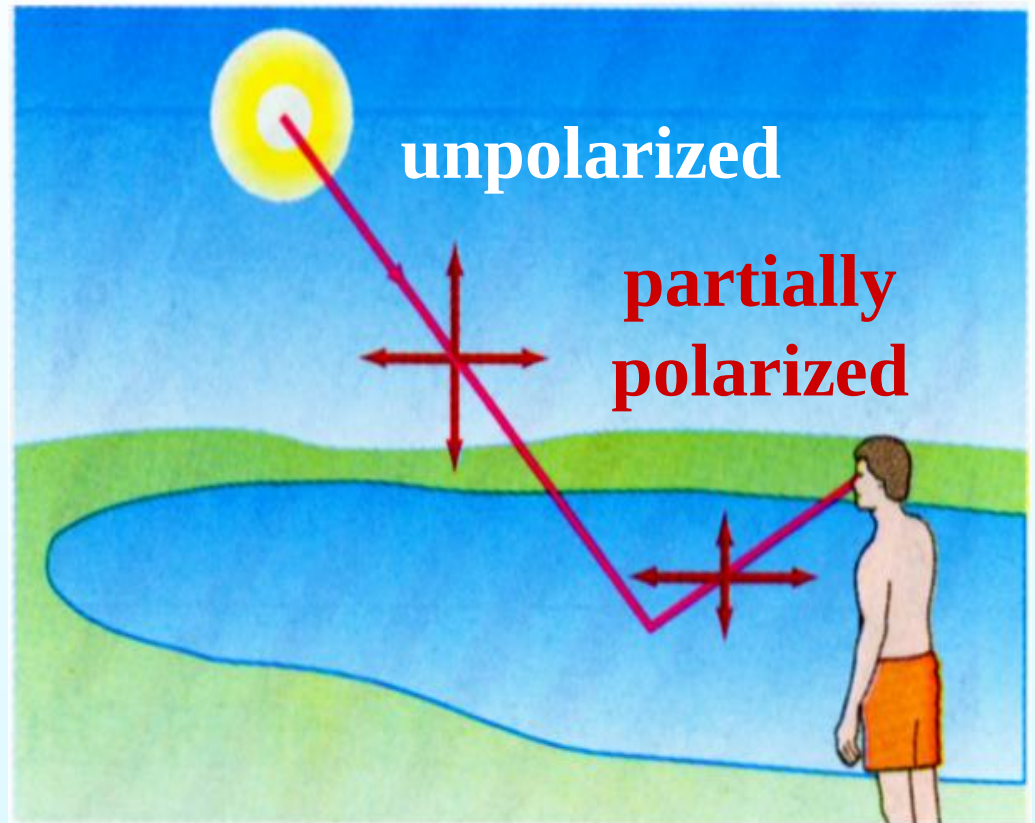
$$I_N = I_0 (\cos^2 \theta)^N = I_0 \cos^{2N} (\pi / N) = 0.90 I_0$$

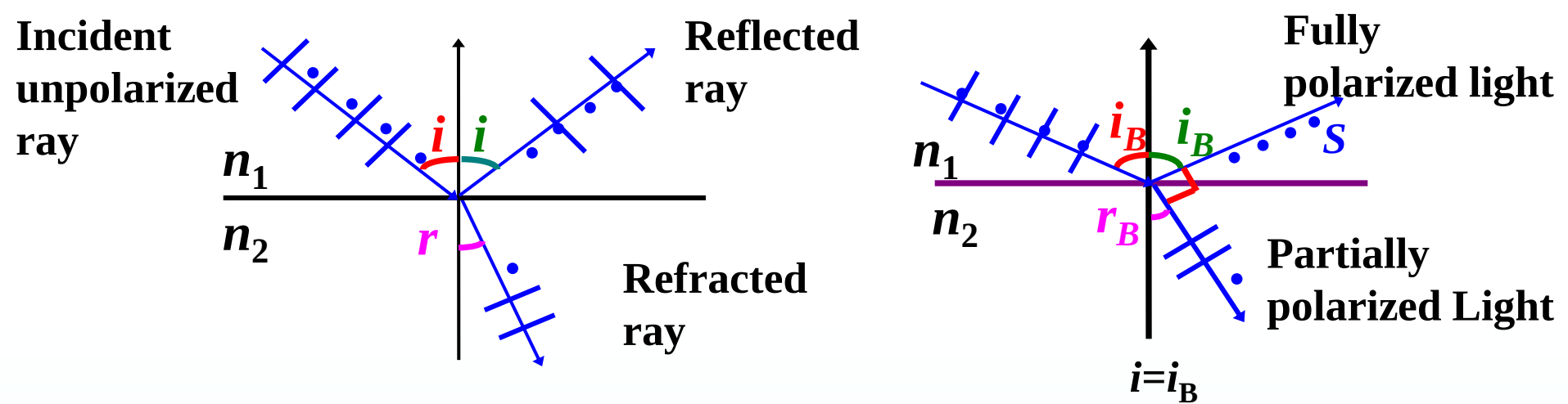
$$N = 24 \quad \theta = 3.75^\circ$$

* 20.3 Polarization by Reflection

Unpolarized light can be polarized, either partially or totally, by **reflection**.

Light reflected from the smooth surface of water in a lake is partially polarized **parallel** to the surface.





When unpolarized light is incident on a reflecting surface, the reflected and refracted beams are partially polarized.

When the light is incident at a particular indecent angle, **Brewster's angle**, the **reflected** light has only **perpendicular** components.

The **refracted** light consists of the original component **parallel** to the page and **weaker** components **perpendicular** to the page.

i_0 — **Brewster's angle** or polarizing angle
(布儒斯特角 或 起偏角)

$$\tan i_B = \frac{n_2}{n_1} = n_{21}$$

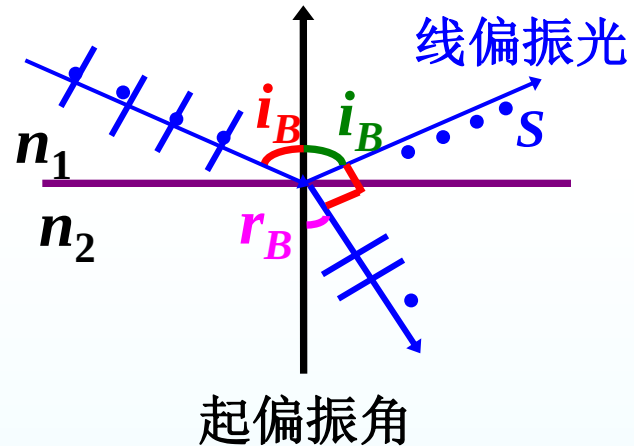
—Brewster's Law

$$\frac{\sin i_B}{\sin r_B} = \frac{n_2}{n_1}$$

$$\tan i_B = \frac{n_2}{n_1} = \frac{\sin i_B}{\cos i_B}$$

$$\cos i_B = \sin r_B = \cos\left(\frac{\pi}{2} - r_B\right)$$

$$i_B + r_B = 90^\circ$$



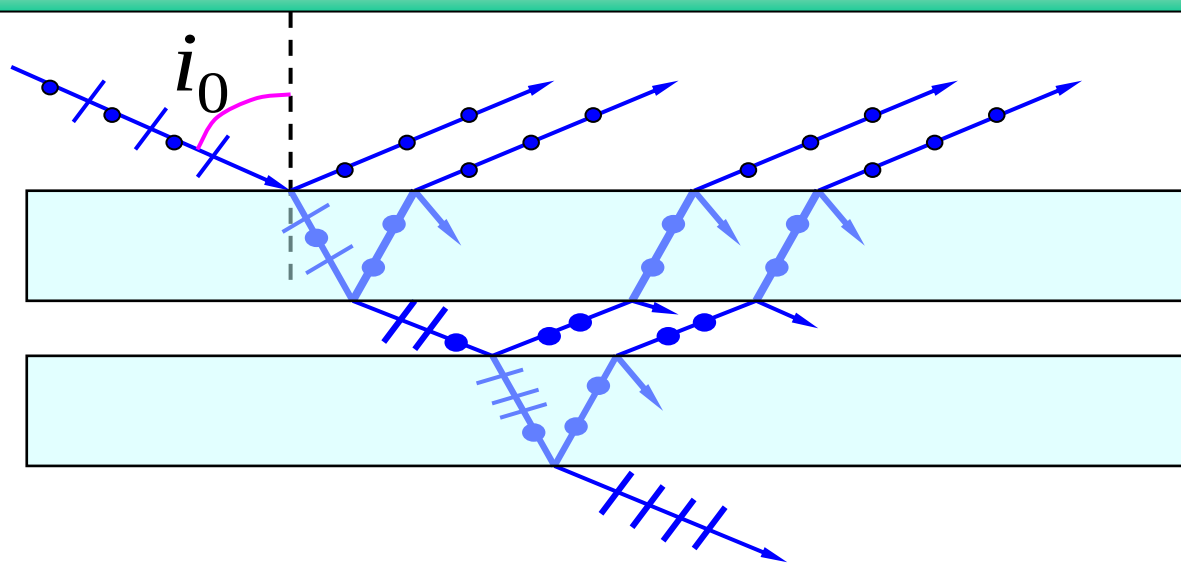
At Brewster's angle, the reflected and refracted rays are *perpendicular*.

Discussion

- (1) 反射光和折射光互相垂直 .
- (2) i_B 是某一个确定的值，而不是某个“界限” 只有当 $i = i_B$ ，反射光才是完全偏振光.
- (3) 无论入射角是多少，折射光总是 “ $| > \cdot$ ” 的部分偏振光，而当 $i \neq i_B$ ，反射光是 “ $\cdot > |$ ” 的部分偏振光.
- (4) 此情形对 $n_2 > n_1$ 或 $n_1 > n_2$ 都成立.

注意 对于一般的光学玻璃，反射光的强度约占入射光强度的7.5%，大部分光将透过玻璃。

利用玻璃片堆产生线偏振光



EXAMPLE:

(a) At what **incident angle** is sunlight reflected from a lake plane-polarized? ($n = 1.33$)

(b) What is the refraction angle?

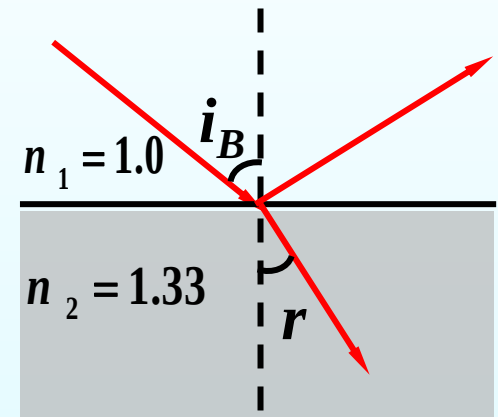
Solution:

$$n = 1.33$$

$$\tan i_0 = 1.33$$

$$i_0 = 53.1^\circ$$

$$r = 90.0^\circ - 53.1^\circ = 36.9^\circ$$



例题：如图，自然光由空气入射到，折射率 $n_2=1.33$ 的水面上，入射角为 i 时使反射光为完全偏振光。今有一块玻璃浸入水中，其折射率 $n_3=1.5$ ，若光由玻璃面反射也成为完全偏振光，求水面与玻璃之间的夹角 α 。

$$\tan r = \frac{n_1}{n_2} = \frac{1}{1.33}$$

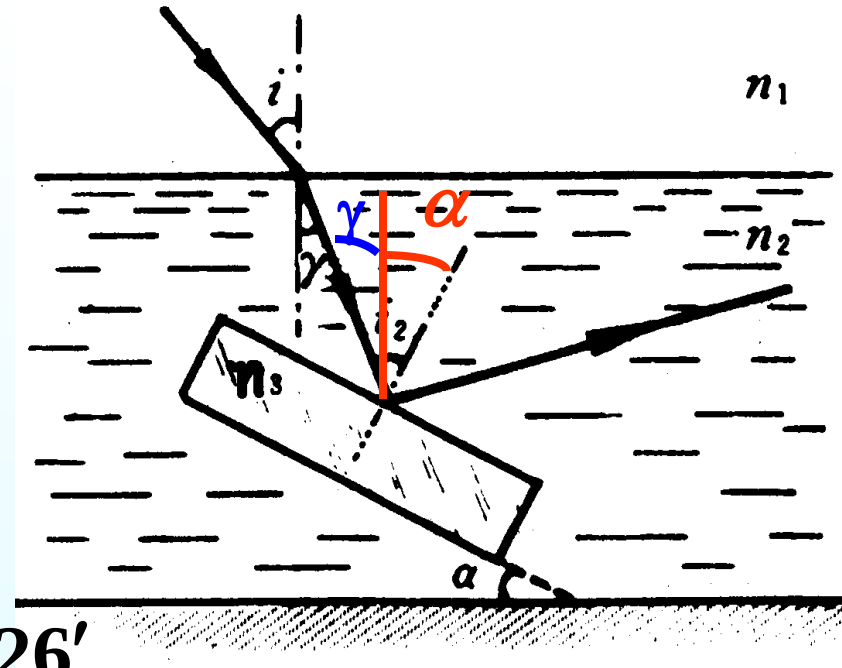
$$\therefore r = 36^\circ 56'$$

又因 i_2 是布儒斯特角，由布儒斯特定律可得：

$$\tan i_2 = \frac{n_3}{n_2} = \frac{1.5}{1.33}$$

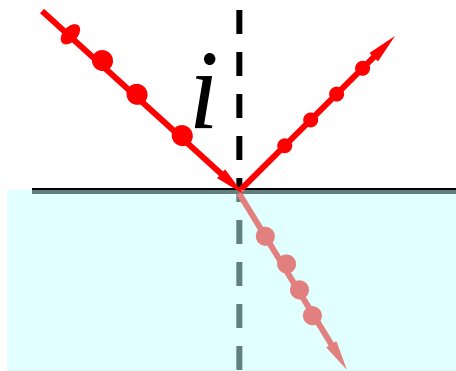
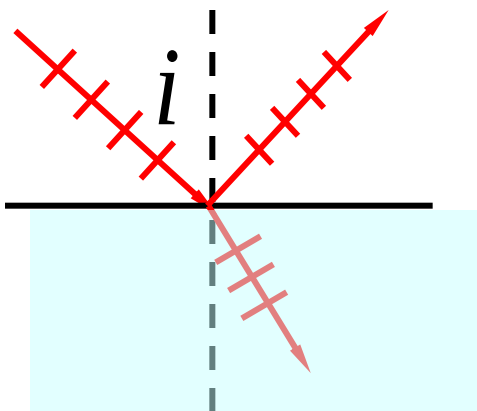
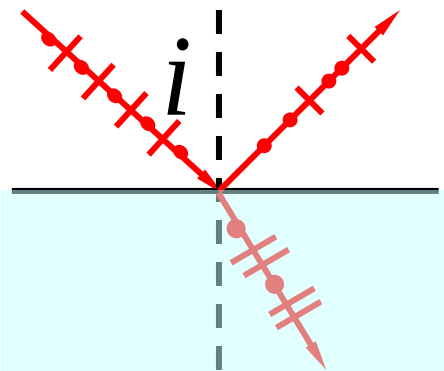
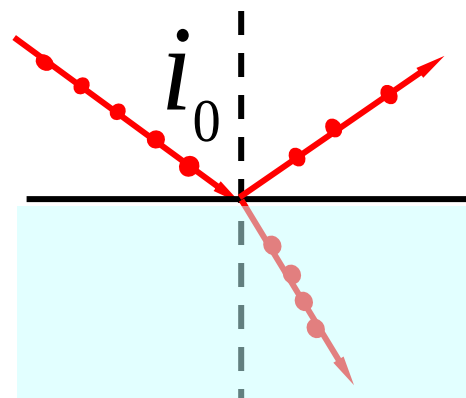
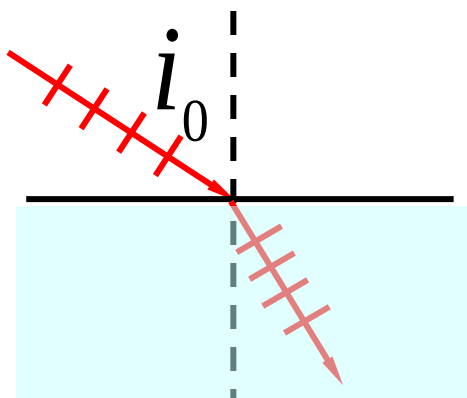
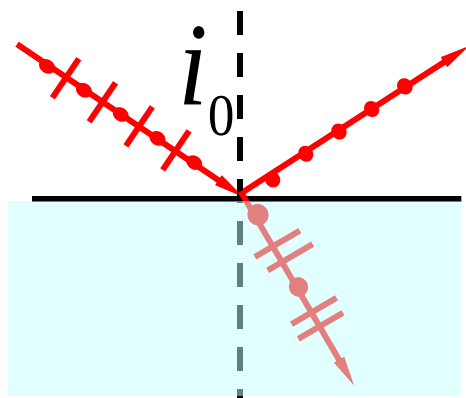
$$\therefore i_2 = 48^\circ 26'$$

$$\therefore \alpha = i_2 - r = 48^\circ 26' - 36^\circ 56' = 11^\circ 30'$$



讨论

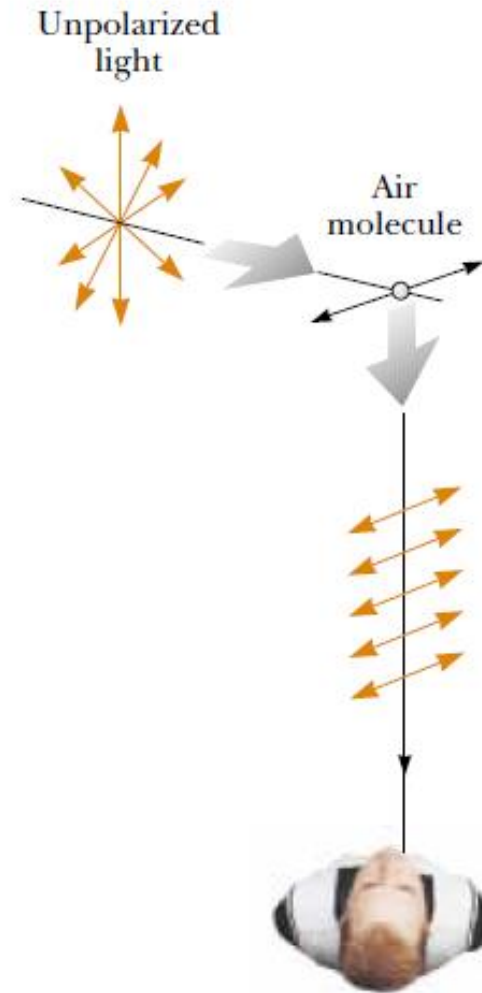
讨论光线的反射和折射(起偏角 i_0)



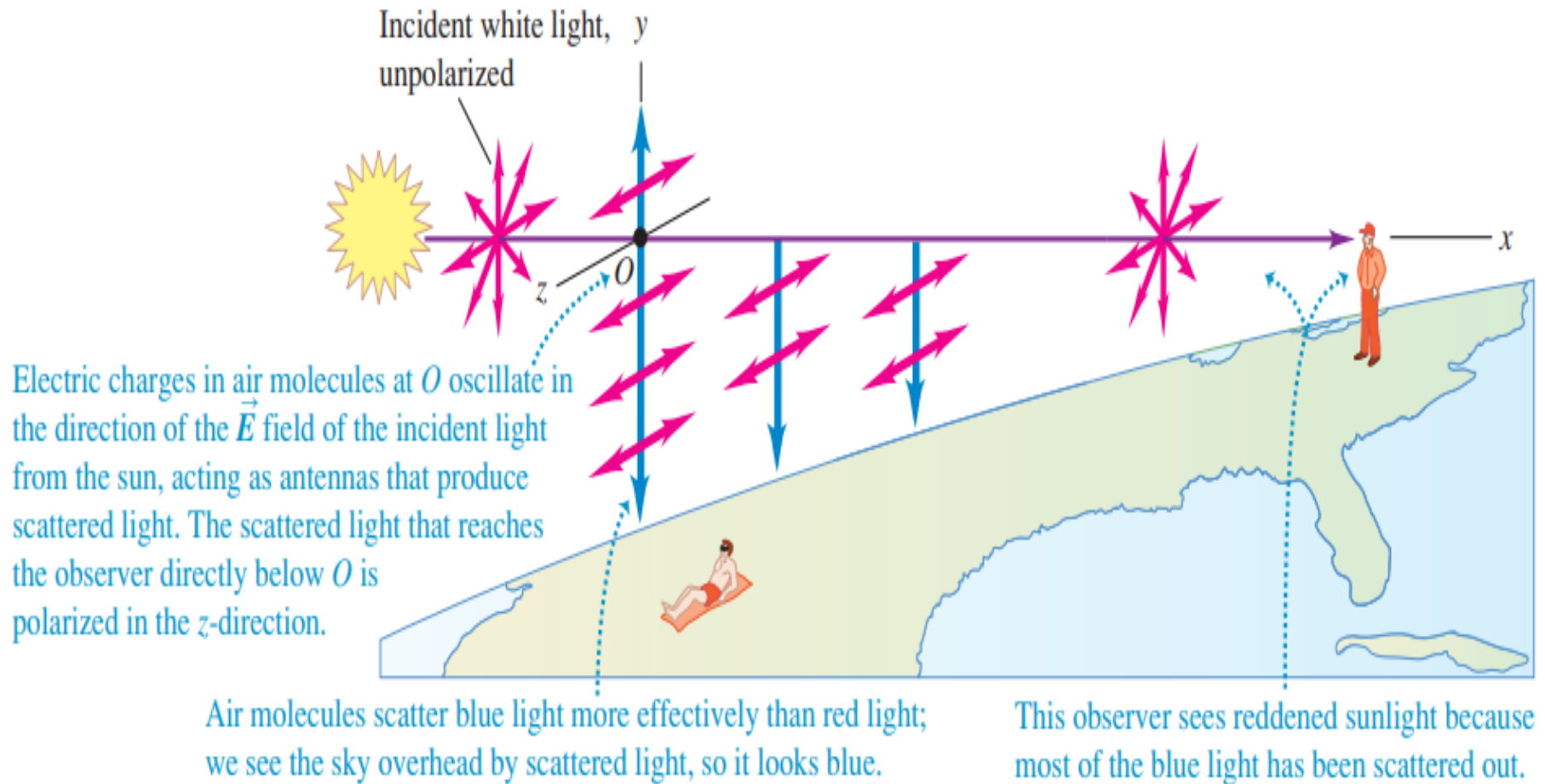
Polarization by Scattering

When light is incident on any material, the **electrons** in the material can **absorb** and **reradiate** part of the light. Such absorption and **reradiation** of light by electrons in the gas molecules that make up air is what causes sunlight reaching an observer on the Earth to be partially polarized.

If the observer in Figure is looking straight up (perpendicular to the original direction of propagation of the light), the vertical oscillations of the charges send no radiation toward the observer.



When the sunbathing observer on the left looks up, he sees **blue, polarized** sunlight that has been scattered by air molecules. The observer on the right sees **reddened, unpolarized** light when he looks at the sun.





Clouds contain a high concentration of water droplets or ice crystals, which also scatter light. Because of this high concentration, light passing through the cloud has many more opportunities for scattering than does light passing through a clear sky. Thus light of **all wavelengths** is eventually **scattered out** of the cloud, so the cloud looks white .



The intensity of the light scattered from air molecules increases in proportion to the fourth power of the frequency and that's why the sky is **blue**.

Near sunset, when sunlight has to travel **a long distance** through the earth's atmosphere, a substantial fraction of the **blue** light is removed by scattering. White light minus blue light looks **yellow** or **red**. This explains the yellow or red hue that we so often see from the setting sun.